Windows Networking Services

❖ Print Server RoleTheory :Overview of the Print Server role.Configuring printers and managing print queues.Practical :Configure a network printer and set up print permissions.Task: Submit a report on the print server setup and test results.

1. Install the Print Server role on your Windows Server and add a new network printer using the Print Management console.

➢ Step 1: Install Print Server role

1. Log in to Windows Server using Administrator.
2. Open Server Manager.
3. Click Manage → Add Roles and Features.
4. Click Next until you reach Server Roles.
5. Expand Print and Document Services.
6. Select Print Server.
7. Click Add Features when prompted.
8. Click Next → Install.
9. Wait until installation completes.

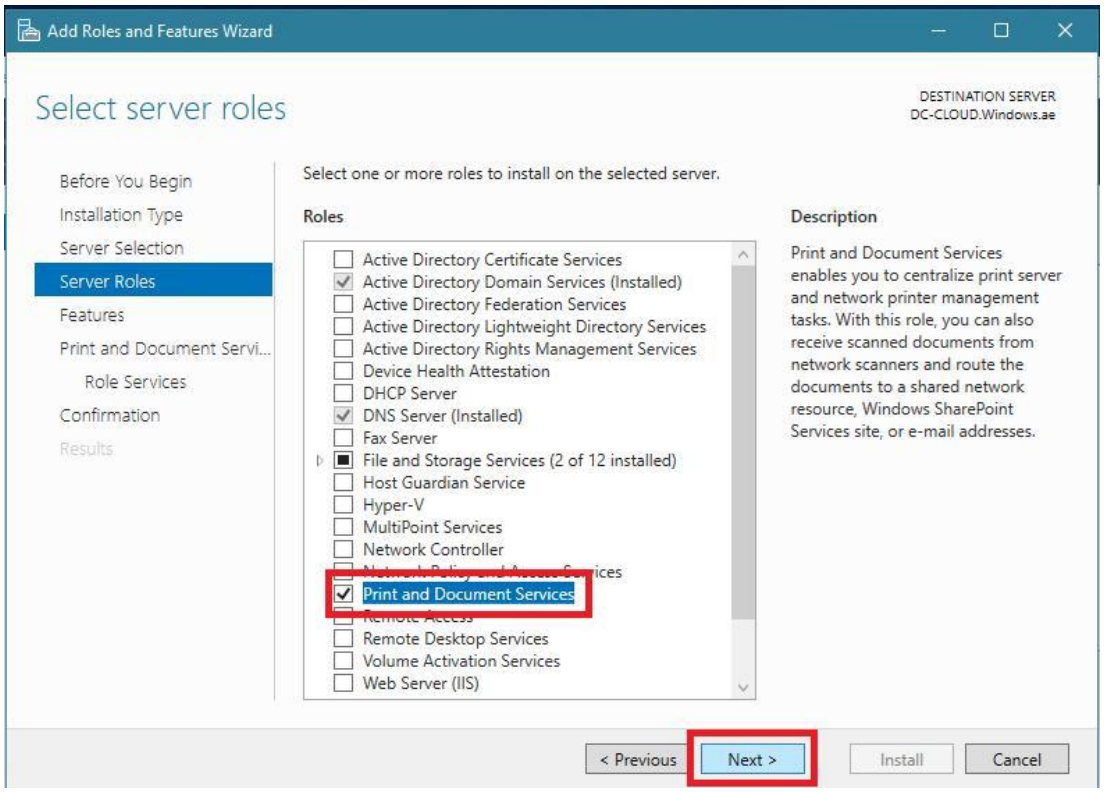
➢ Step 2: Open Print Management

1. Open Server Manager.
2. Click Tools → Print Management.
3. Expand:
   - Print Management
   - Print Servers
   - Your Server Name
4. Right-click Printers.
5. Select Add Printer.

➢ Step 3: Add a network printer

For a physical network printer:

1. Select Add a new printer using an existing port if its TCP/IP port already exists, or create a new Standard TCP/IP Port.
2. Enter the printer's IP address.
3. Install the appropriate printer driver.
4. Set the printer name to:
   CollegeFestPrinter
5. Select the option to share this printer.
6. Set the share name to:
   CollegeFestPrinter
7. Complete the wizard.
   Example network path:



2. Configure print permissions for a shared network printer so that only specific users (create two test user accounts) can print, while others are denied access.

Hint: Use the Security tab in the printer's Properties to set Allow/Deny permissions for each user.

- **Step 1: Create two test users**
  1. Open Active Directory Users and Computers.
  2. Open your domain.
  3. Right-click the appropriate OU.
  4. Select New → User.
  5. Create these accounts:

Username	Purpose
PrintAllowed1	Allowed to print
PrintDenied1	Denied printing

  6. Set a strong, unique password for each account.
  7. Complete the user creation wizard.

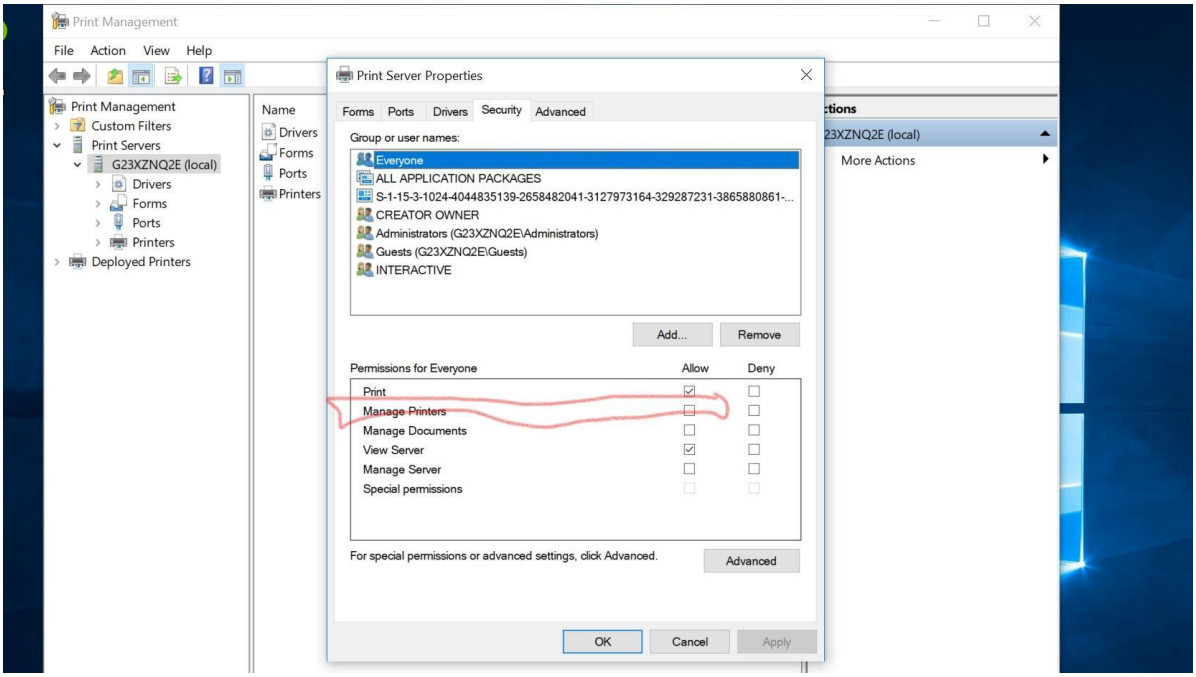
- **Step 2: Open printer security settings**
  1. Open Print Management.
  2. Expand Print Servers → Your Server → Printers.
  3. Right-click CollegeFestPrinter.
  4. Select Properties.
  5. Open the Security tab.

- **Step 3: Configure allowed user**
  1. Click Add.
  2. Enter PrintAllowed1.
  3. Click Check Names → OK.
  4. Select PrintAllowed1.
  5. Set the permissions:

Permission	Allow	Deny
Print	Yes	No
Manage this printer	No	No
Manage documents	No	No

- **Step 4: Configure denied user**
  1. Click Add.
  2. Enter PrintDenied1.
  3. Click Check Names → OK.
  4. Select PrintDenied1.
  5. Set:

Permission	Allow	Deny
Print	No	Yes
Manage this printer	No	No
Manage documents	No	No



3. Send a test print job from both an allowed and a denied user account, then use the Print Management console to monitor and manage the print queue, noting what happens for each user.

- **Scenario A: PrintAllowed1 – Successfully print**
  1. Log in to the client PC using PrintAllowed1.
  2. Press Win + R.
  3. Enter the shared printer path:
     \\192.168.1.10\CollegeFestPrinter
  4. Connect to the shared printer.
  5. Open Notepad.
  6. Type:
     College Fest - Event Poster
     Welcome to Our College Fest!
  7. Click File → Print.
  8. Select CollegeFestPrinter.
  9. Click Print.
     - Expected Result: Print Successful
     - Observation: PrintAllowed1 can connect to the shared network printer and submit a print job successfully.
     - Reason: The user has Print permission in the printer's Security settings.

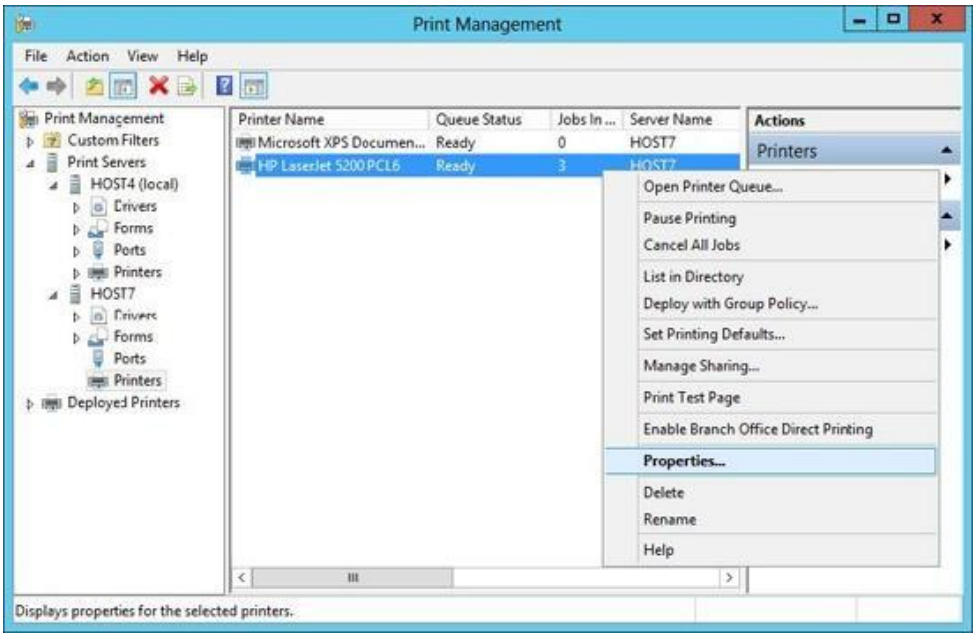
- **Scenario B: PrintDenied1 – Printing denied**
  1. Log out from PrintAllowed1.
  2. Log in using PrintDenied1.
  3. Open Run (Win + R).
  4. Enter:
     \\192.168.1.10\CollegeFestPrinter
  5. Try connecting to the printer.
  6. Alternatively, select the printer in Notepad and attempt to print.
     - Expected Result: Access Denied
     - Observation: PrintDenied1 cannot connect to or submit documents to the shared printer. Windows displays an access or permission error.
     - Reason: The user has an explicit Deny permission for Print.

- **Step 3: Monitor the print queue**

  On Windows Server:
  1. Open Print Management.
  2. Expand Print Servers → Your Server → Printers.
  3. Right-click CollegeFestPrinter.
  4. Select Open Printer Queue.
  5. Check the submitted print jobs.

  Expected results:

User	Result	Print queue
PrintAllowed1	Successful	Job appears
PrintDenied1	Access Denied	No accepted job appears



4. Change the priority of two print jobs in the print queue so that one user's job prints before another's, regardless of submission order.

Hint: Right-click the job in the print queue and adjust the priority value.

   - Objective: Make one user's print job print before another user's job, regardless of submission order.

User	Document	Priority
PrintAllowed1	EventPoster.pdf	99
EventManager1	TicketDesign.pdf	1

   A higher priority number means a higher printing priority. Windows supports print job priorities from 1 to 99.
   Microsoft Learn+1

   - **Step 1: Submit two print jobs**
     1. Log in as PrintAllowed1.
     2. Submit a print job named EventPoster.pdf.
     3. Log in as EventManager1.
     4. Submit another job named TicketDesign.pdf.
        For easier testing, pause the printer before submitting both jobs, so they remain in the queue.

   - **Step 2: Open print queue**
     1. Return to Windows Server.
     2. Open Print Management.
     3. Navigate to:
        Print Management
        → Print Servers
        → Your Server

→ Printers
→ CollegeFestPrinter

4. Right-click the printer and select Open Printer Queue.

- **Step 3: Set priority for EventPoster.pdf**
  1. Right-click EventPoster.pdf.
  2. Select Properties.
  3. Open the General tab.
  4. Find the Priority slider.
  5. Set the value to 99.
  6. Click Apply → OK.

- **Step 4: Set priority for TicketDesign.pdf**
  1. Right-click TicketDesign.pdf.
  2. Select Properties.
  3. Open the General tab.
  4. Set the Priority value to 1.
  5. Click Apply → OK.

5. Simulate a real-world scenario: Imagine your college fest team is printing event posters. Configure the printer so that 'Event Managers' group can manage documents (pause, resume, cancel), while 'Volunteers' can only print, and test these permissions by logging in as both roles.

- Real-world scenario: Your college fest team needs to print event posters, tickets and other promotional materials.

- The Event Managers should be able to manage all documents in the print queue, while Volunteers should only be able to print.

- **Step 1: Create two security groups**
  1. Open Active Directory Users and Computers.
  2. Open your domain.
  3. Right-click your desired OU.
  4. Select New → Group.

Create these two groups:

Group name	Group scope	Group type
Event Managers	Global	Security
Volunteers	Global	Security

- **Step 2: Create test user accounts**

Create the following users:

Username	Role
EventManager1	Event Manager
Volunteer1	Volunteer

Set strong passwords and complete the account creation wizard.

- **Step 3: Add users to their groups**

  Add EventManager1:

  1. Double-click Event Managers.
  2. Open Members.
  3. Click Add.
  4. Enter EventManager1.
  5. Click Check Names → OK → Apply.

  Add Volunteer1:

  1. Double-click Volunteers.
  2. Open Members.
  3. Click Add.
  4. Enter Volunteer1.
  5. Click Check Names → OK → Apply.

- **Step 4: Configure printer permissions**

  1. Open Print Management.
  2. Expand Print Servers → Your Server → Printers.
  3. Right-click CollegeFestPrinter.
  4. Select Properties.
  5. Open the Security tab.

  Remove broad groups that grant unintended Print access, such as Everyone, where applicable.
  Now add Event Managers and Volunteers.

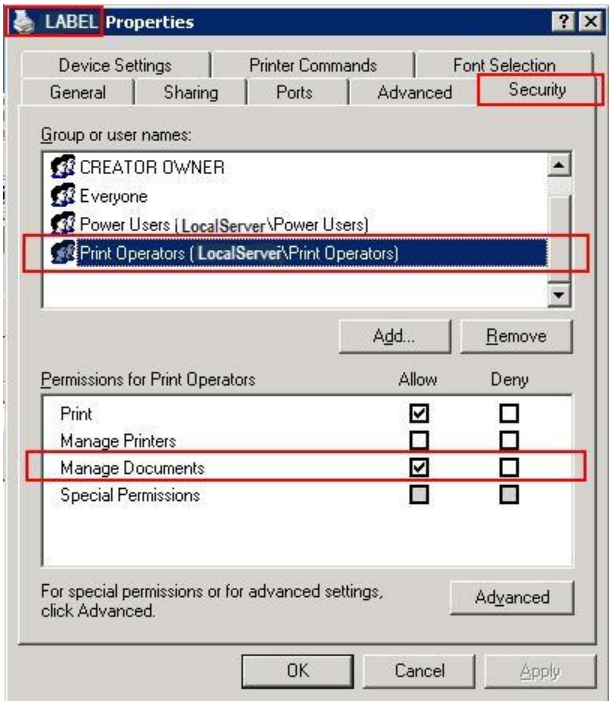
Configure Event Managers:
Select the Event Managers group and configure:

Permission	Allow
Print	Enabled
Manage this printer	Disabled
Manage documents	Enabled

Configure Volunteers:

Select the Volunteers group and configure:

Permission	Allow
Print	Enabled
Manage this printer	Disabled
Manage documents	Disabled


LABEL Properties
Device Settings
Printer Commands
Font Selection
General
Sharing
Ports
Advanced
Security
Group or user names:
CREATOR OWNER
Everyone
Power Users (LocalServer\Power Users)
Print Operators (LocalServer\Print Operators)
Add...
Remove
Permissions for Print Operators
Allow
Deny
Print
Manage Printers
Manage Documents
Special Permissions
For special permissions or for advanced settings, click Advanced.
Advanced
OK
Cancel
Apply

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